Optimal Stopping with Memory: Deep Recurrent Policies for American Options under Rough Volatility and Multi-Market Backtesting

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1 Introduction

Pricing an American option means deciding when to exercise it, which makes it an *optimal stopping* problem. The classical numerical methods (binomial trees, the least-squares Monte Carlo method of Longstaff and Schwartz [3] and Deep Optimal Stopping [2]) all solve it by backward induction. They assume that the price process is Markovian, so that today's state is enough to decide.

Real markets are not Markovian in the price alone: volatility is persistent and, empirically, *rough* [4]. When the past matters, backward induction needs an ever larger state, or it carries approximation errors from one date into the previous one. Damera Venkata and Bhattacharyya [1] proposed the *Optimal Stopping Policy Gradient* (OSPG), which trains a recurrent network to choose stopping times directly and needs no backward recursion.

We (i) replicate OSPG on the paper's Markovian option benchmarks, (ii) price American puts under the rough Bergomi model, where the price is non-Markovian, and (iii) backtest learned exit-timing rules on market data, comparing each with the classical methods.

2 Optimal stopping and American options

The problem. Let $(S_j)_{j=0}^{H}$ be a price process observed at dates $t_0 < \cdots < t_H$, with filtration $(\mathcal{F}_j)$ generated by its history. Stopping at date j pays a reward g_j that is known at date j; for an American (Bermudan) put under the risk-neutral measure $\mathbb{Q}$, $g_j = e^{-rt_j}(K - S_j)^+$. A *stopping time* is a random date τ with $\{\tau = j\} \in \mathcal{F}_j$: the decision to stop uses only the past. The price is

$$V_0 = \sup_{\tau \in \mathcal{T}_{0,H}} \mathbb{E}^{\mathbb{Q}}\big[g_\tau\big]. \tag{1}$$

Snell envelope. Define backwards

$$U_H = g_H, \qquad U_j = \max\big(g_j,\ \mathbb{E}[U_{j+1} \mid \mathcal{F}_j]\big), \quad j = H-1, \ldots, 0. \tag{2}$$

The process (U_j) is the smallest supermartingale that dominates (g_j); the stopping time $\tau^* = \min\{j : U_j = g_j\}$ is optimal and $V_0 = U_0$ [6]. The term $C_j = \mathbb{E}[U_{j+1} \mid \mathcal{F}_j]$ is the *continuation value*: one exercises as soon as the immediate reward is at least the value of waiting.

The Markov case. If S is Markov and $g_j = G_j(S_j)$, the continuation value is a function of today's state only, $C_j = c_j(S_j)$. The binomial tree computes c_j exactly on a lattice; Longstaff and Schwartz estimate it by regressing realised discounted cash flows on functions of S_j [3]; Deep Optimal Stopping learns the exercise decision at each date with a neural network of S_j [2]. When the dynamics have memory (as under rough volatility), C_j depends on the whole path $(S_0, \ldots, S_j)$ and these methods lose information.

Continuous time: a free-boundary problem. When exercise is allowed at all times, a put is exercised when S_t falls below a boundary $L(t)$. Above it the value solves the Black–Scholes equation $\partial_t V + rS\partial_S V + \frac{1}{2}\sigma^2 S^2 \partial_{SS} V - rV = 0$; below it $V = K - S$; and at $S = L(t)$ the value meets the payoff smoothly, $\partial_S V = -1$ (smooth pasting). The boundary is unknown in advance. For a perpetual put it is constant [6],

$$L^* = \frac{2rK}{2r + \sigma^2}, \qquad V(S) = (K - L^*)\Big(\frac{S}{L^*}\Big)^{-2r/\sigma^2} \quad (S > L^*); \tag{3}$$

with $K = 40$, $r = 0.06$, $\sigma = 0.2$ this gives $L^* = 30$ and $V(36) = 5.79$.

Solving the finite-maturity problem numerically. For a finite maturity there is no closed form, and the free-boundary problem is written as a linear complementarity problem: with $\mathcal{L}V = (r - \delta)S\partial_S V + \frac{1}{2}\sigma^2 S^2 \partial_{SS} V - rV$,

$$\min\big(-\partial_t V - \mathcal{L}V,\ V - g\big) = 0, \qquad V(T, S) = g(S). \tag{4}$$

Where waiting is optimal the first term vanishes, which is the Feynman–Kac equation for the discounted expected payoff; where exercise is optimal $V = g$. We solve it in $x = \log S$, where the coefficients are constant, with Crank–Nicolson time steps (the first four fully implicit, to damp the oscillations caused by the kink of the payoff) and impose $V \geq g$ by a penalty iteration [5]: each tridiagonal system is solved with a large penalty on the nodes where $V < g$, until those nodes stop changing. A Bermudan option is priced by solving the European equation between exercise dates and taking $\max(V, g)$ on each of them. Table 1 compares the result with a 5,000-step-per-year binomial tree: the two independent methods agree to within 0.0004 on all twenty cases, so either can serve as the ground truth for the simulation methods. The early-exercise premium is the gap between the American and European columns, and the Bermudan price with 50 exercise dates per year is only about 0.007 below the American one. For $S_0 = 36$, $\sigma = 0.2$, $T = 1$ the computed boundary rises from $L(0) = 32.9$ to $L(T^-) = K = 40$ and stays above the perpetual boundary $L^* = 30$ throughout: with less time left, waiting is worth less, so exercise starts at higher prices.

Optional stopping. If (M_j) is a martingale and τ a bounded stopping time, then $\mathbb{E}[M_\tau] = M_0$: no stopping rule can improve on a fair game. Two consequences guide our experiments: stopping a standard Brownian motion is worth exactly 0, so memory can only help when increments are dependent; and if market prices are close to martingales, no exit rule can beat holding to the horizon in mean, so any benefit of memory should appear mainly in risk-adjusted terms.

3 Method: optimal stopping policy gradients

Stochastic stopping policy. A network with parameters θ reads the history and outputs, at each date, $\varphi_j^\theta = \mathbb{P}(\text{stop at } j \mid \text{not stopped before, } \mathcal{F}_j)$, with $\varphi_H = 1$. Because stopping does not change the path, the distribution of the stopping time on a given path is exact [1]:

$$\psi_j^\theta = \mathbb{P}(\tau = j) = \varphi_j^\theta \prod_{n<j} \big(1 - \varphi_n^\theta\big), \qquad \psi_H^\theta = \prod_{n<H} \big(1 - \varphi_n^\theta\big). \tag{5}$$

The expected reward $J(\theta) = \mathbb{E}\big[\sum_j \psi_j^\theta g_j\big]$ can therefore be evaluated on pre-simulated paths, without rollouts and without backward recursion.

Gradient and loss. By the log-derivative trick,

$$\nabla_\theta J = \mathbb{E}\Big[\sum_{j=0}^{H} (g_j - b)\, \psi_j^\theta\, \nabla_\theta \log \psi_j^\theta\Big], \tag{6}$$

Table 1: American put, $K = 40$, $r = 0.06$: European (Black–Scholes), Bermudan (50 exercise dates per year) and American prices from the binomial tree and the finite-difference solver of the free-boundary problem.

S_0	σ	T	European	Bermudan		American	
				tree	PDE	tree	PDE
36	0.2	1	3.844	4.478	4.478	4.487	4.487
36	0.2	2	3.763	4.840	4.840	4.848	4.848
36	0.4	1	6.711	7.101	7.101	7.109	7.109
36	0.4	2	7.700	8.507	8.507	8.514	8.514
38	0.2	1	2.852	3.250	3.250	3.257	3.257
38	0.2	2	2.991	3.745	3.745	3.751	3.751
38	0.4	1	5.834	6.148	6.148	6.155	6.154
38	0.4	2	6.979	7.668	7.668	7.675	7.675
40	0.2	1	2.066	2.314	2.314	2.320	2.319
40	0.2	2	2.356	2.885	2.884	2.890	2.890
40	0.4	1	5.060	5.312	5.312	5.318	5.318
40	0.4	2	6.326	6.917	6.917	6.923	6.923
42	0.2	1	1.465	1.617	1.617	1.621	1.621
42	0.2	2	1.841	2.212	2.212	2.217	2.217
42	0.4	1	4.379	4.583	4.582	4.588	4.588
42	0.4	2	5.736	6.244	6.244	6.250	6.250
44	0.2	1	1.017	1.110	1.110	1.113	1.113
44	0.2	2	1.429	1.690	1.690	1.693	1.693
44	0.4	1	3.783	3.948	3.948	3.953	3.953
44	0.4	2	5.202	5.641	5.641	5.647	5.647

for any constant b, since $\sum_j \psi_j = 1$ makes the b term vanish. We minimise the temporal loss $\mathcal{L} = -\frac{1}{N}\sum_{i,j} \mathrm{sg}\big[\psi_{ij}(g_{ij} - b)\big] \log \psi_{ij}$, where sg stops gradients, whose gradient is exactly $-\nabla_\theta J$ (checked numerically in our tests). The update raises φ_j when the ratio of the immediate value $\psi_j g_j$ to the continuation value $\sum_{n>j} \psi_n g_n$ exceeds the current odds $\varphi_j/(1-\varphi_j)$, an implicit Snell-envelope comparison at every date.

Architecture and training. The recurrent policy (OSPG-GRU) is a single-layer GRU with 20 hidden units followed by a linear layer and a sigmoid; the memoryless policy (OSPG-MLP) applies a two-layer network with 20 units to each date's inputs, with batch normalisation of its input and of each hidden layer's output, as in the paper; at decision time the normalisation uses the statistics from training, so a path's decision never depends on the other paths. Inputs at date j are the standardised features, the current reward and j/H, all known at date j. We train with Adam (learning rate 10^{-3}), keep the parameters with the best validation value, and stop at the first date with $\varphi_j \geq \frac{1}{2}$ at test time. Three implementation choices were needed. The output bias starts at -2, because a policy started near $\varphi = \frac{1}{2}$ converged to "stop immediately" on the in-the-money put. A small penalty $10^{-3}\,\overline{z^2}$ on the pre-sigmoid outputs keeps φ away from 0: the policy gradient is proportional to φ, and without the penalty the GRU policy stalled at "never stop" on the 50-asset max-call. Finally, date 0, where all simulated paths coincide, is decided by comparing g_0 with the value achieved by the trained policy, as for the baselines.

Baselines. Longstaff–Schwartz regression on $[1, x, x^2]$ of the standardised features and current reward (in-the-money paths only for options); Deep Optimal Stopping with one network per date, warm-started from the next date; and fitted Q-iteration with one network (MLP or GRU) for all dates [1]. The baseline networks are run without batch normalisation: on the American put it lowered FQI-MLP from 4.480 and 4.462 to 4.426 and 4.442 (two seeds), so we kept the stronger configuration for the baselines, which can only make the comparison harder for OSPG. Every method sees the same features, and all values are measured out of sample, so they are lower bounds on the

true price; the binomial tree and the finite-difference solver of Section 2 give the exact prices they are measured against.

4 Experiments and results

4.1 Replication of the option benchmarks

Set-up. We ran three Markovian problems for which the exact price is known. *E1*: the Bermudan put of Section 2 ($S_0 = 36$, $K = 40$, $r = 0.06$, $\sigma = 0.2$, $T = 1$, 50 exercise dates; exact price 4.478 from the tree and the PDE solver). *E2*: the paper's Bermudan max-call on $d = 20$ and 50 independent assets ($s_0 = 100$, 9 exercise dates, Table 1 of [1]); the reference is the midpoint of the lower and upper bounds of Becker et al. [2], $[51.571, 51.765]$ and $[69.582, 69.889]$. *E3*: the paper's American call on the geometric average of $d = 7$ and 20 correlated assets (100 exercise dates, Appendix D.3); the geometric average of correlated geometric Brownian motions is itself one, so the exact price follows from a one-dimensional tree. Each problem was simulated with 100,000 paths (40,000 for E2), split at random into 50% test and 50% training, of which a fifth is held out for validation, and repeated for three seeds. Every method sees the same features and is scored on the same test paths.

Table 2: Out-of-sample values, mean over three seeds (standard deviation across seeds). The standard error of one test set is about 0.013 (put), 0.16 (max-call) and 0.065 (geometric call). DOS was not run on the 100-date geometric call.

Method	Put	Max-call d=20	Max-call d=50	Geo-call d=7	Geo-call d=20
LSM	4.458 (0.004)	51.69 (0.20)	69.83 (0.32)	10.169 (0.012)	9.899 (0.011)
DOS	4.465 (0.007)	51.69 (0.16)	69.72 (0.29)	–	–
FQI-MLP	4.467 (0.011)	51.68 (0.12)	69.67 (0.21)	10.145 (0.068)	9.874 (0.056)
FQI-GRU	–	51.73 (0.14)	69.81 (0.37)	–	–
OSPG-MLP	4.473 (0.008)	51.80 (0.20)	69.93 (0.28)	10.255 (0.046)	9.989 (0.017)
OSPG-GRU	4.465 (0.015)	51.72 (0.21)	69.87 (0.25)	10.237 (0.065)	9.941 (0.030)
Reference	4.478	51.67	69.74	10.260	10.033
Paper, DNN-OSPG	–	51.02	69.06	10.252	10.010

Every method recovers the price; OSPG-MLP comes closest. Table 2 shows all methods within 1.6% of the exact price. The memoryless OSPG policy is the most accurate on the three problems where the differences are larger than the noise: -0.12% on the put, and -0.05% and -0.44% on the geometric call, against -0.9% to -1.6% for Longstaff–Schwartz and fitted Q-iteration. This is the paper's ranking (OSPG ahead of FQI and LS), and our OSPG errors are of the same size as the paper's (0.07% and 0.23%), although on the geometric call we trained on about thirty times fewer paths. On the max-call every method is within one standard error of the reference. OSPG-MLP's 51.80 and 69.93 lie slightly above Becker et al.'s upper bounds, but a mean of three test sets has a standard error of about 0.09, so neither is significant.

Paired comparisons. Because all methods are evaluated on the same test paths, we compare them path by path: the noise common to both methods cancels and much smaller differences become visible (`results/paired.md`). OSPG-GRU is ahead of Longstaff–Schwartz and FQI-MLP on both geometric calls by 0.04 to 0.09 (z between 2.6 and 4.6), and of DOS and FQI-MLP on the 50-asset max-call by 0.15 and 0.20 ($z = 3.9$ and 5.4). It is, however, behind OSPG-MLP on the put (0.008, $z = 3.0$), on both max-calls (0.086 and 0.064, $z = 2.9$ and 2.2) and on the 20-asset geometric call (0.049, $z = 3.7$).

Memory has a small cost when there is nothing to remember. In these Markovian problems the history carries no information beyond today's state, so the recurrent policy has to learn

to ignore it, and it does so imperfectly: it loses 0.1% to 0.5% against the memoryless policy and trains 1.3 to 4 times more slowly. Part of the gap may come from batch normalisation, which the MLP uses (as in the paper) and the GRU does not. The loss is small next to the margin by which both OSPG policies beat the backward-induction methods on the harder problems, so the recurrent policy remains a reasonable default when it is not known whether the problem has memory.

The paper's own max-call values. The paper reports 50.86–51.02 for $d = 20$ and 68.87–69.06 for $d = 50$, about 0.8 below Becker et al.'s lower bounds and below every method here, including our implementations of the paper's baselines. We therefore compare with Becker et al. rather than with the paper's numbers; the paper's conclusions about the max-call (all methods within a few tenths of each other) are unaffected.

4.2 Fractional Brownian motion and rough volatility

Fractional Brownian motion. For a Hurst parameter $h \in (0, 1)$, fractional Brownian motion (fBm) W^h is a centred Gaussian process with covariance

$$\mathbb{E}[W_t^h W_s^h] = \frac{1}{2}\left(t^{2h} + s^{2h} - |t - s|^{2h}\right). \tag{7}$$

When $h = 1/2$, fBm is standard Brownian motion. For $h < 1/2$ its increments are negatively correlated, while for $h > 1/2$ they are positively correlated. For $h \neq 1/2$, fBm is not a semimartingale, so the usual Markovian state representation is not available.

The simulator generates exact finite-dimensional Gaussian samples on the chosen time grid. It constructs the covariance matrix C from (7), computes its Cholesky factorisation $C = LL^\top$, draws independent standard normal vectors Z, and sets $W = ZL^\top$. The deterministic initial value $W_0^h = 0$ is then prepended to every path. As a numerical check, for $h = 0.3$, 40,000 paths and 10 time intervals, the empirical terminal variance was 1.0022 against the theoretical value 1.0000; the maximum absolute difference between the empirical and theoretical covariance matrices was 0.00792.

fBm stopping experiment. Following the stopping problem in the OSPG paper [1], the reward is the fBm value itself and the objective is

$$\sup_{0 \leq \tau \leq H} \mathbb{E}[W_\tau^h]. \tag{8}$$

The observation supplied to a policy contains the current fBm value and normalised time; the reward at date j is $R_j = W_{t_j}^h$. We tested $h \in \{0.05, 0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 0.95\}$ using 20,000 paths, 100 time intervals and two random 50/50 train–test splits, with 20% of the training portion reserved for validation. The methods were DOS, FQI-MLP, FQI-GRU, OSPG-MLP and OSPG-GRU. This is a reduced-compute reproduction; the paper used 40,000 paths and ten splits.

At $h = 0.5$, all five learned-method estimates are close to zero, as expected for a bounded stopping rule applied to a Brownian martingale. Away from $h = 0.5$, the attainable mean reward is positive. In this reduced experiment, the recurrent models are competitive, but the GRU variants do not dominate the MLP variants at every h. The results therefore support the role of increment dependence in the stopping problem, but do not establish a uniform GRU advantage. The two-split sample is small, so differences between methods should be interpreted as descriptive rather than as conclusive statistical rankings.

Table 3: fBm stopping results. Entries are mean test rewards; parentheses show the standard deviation across the two split-level estimates.

h	DOS	FQI-MLP	FQI-GRU	OSPG-MLP	OSPG-GRU
0.05	0.700 (0.005)	0.910 (0.017)	1.297 (0.001)	1.153 (0.004)	1.286 (0.004)
0.10	0.508 (0.024)	0.740 (0.004)	1.040 (0.004)	0.926 (0.003)	1.032 (0.008)
0.20	0.304 (0.002)	0.481 (0.009)	0.643 (0.009)	0.578 (0.004)	0.639 (0.002)
0.30	0.187 (0.007)	0.299 (0.008)	0.366 (0.005)	0.334 (0.001)	0.362 (0.008)
0.40	0.092 (0.021)	0.132 (0.008)	0.158 (0.012)	0.141 (0.010)	0.151 (0.015)
0.50	-0.001 (0.007)	-0.002 (0.009)	-0.006 (0.003)	-0.004 (0.006)	-0.008 (0.011)
0.60	0.070 (0.003)	0.081 (0.013)	0.099 (0.010)	0.096 (0.003)	0.094 (0.007)
0.70	0.165 (0.002)	0.177 (0.000)	0.202 (0.001)	0.191 (0.002)	0.193 (0.002)
0.80	0.245 (0.005)	0.258 (0.004)	0.260 (0.001)	0.266 (0.004)	0.262 (0.002)
0.90	0.302 (0.007)	0.306 (0.003)	0.282 (0.013)	0.314 (0.008)	0.306 (0.007)
0.95	0.353 (0.001)	0.346 (0.004)	0.328 (0.028)	0.357 (0.002)	0.349 (0.001)

Rough Bergomi model and hybrid simulation. For the rough-volatility experiments we use the rough Bergomi model

$$dS_t = rS_t\,dt + \sqrt{v_t}S_t\,dZ_t, \tag{9}$$

$$Z_t = \rho W_t + \sqrt{1-\rho^2}\,W_t^{\perp}, \tag{10}$$

$$v_t = \xi_0 \exp\left(\eta\widetilde{W}_t^h - \frac{\eta^2}{2}t^{2h}\right), \tag{11}$$

$$\widetilde{W}_t^h = \sqrt{2h}\int_0^t (t-s)^{h-1/2}\,dW_s. \tag{12}$$

Here W and $W^{\perp}$ are independent Brownian motions, and ρ controls the correlation between the stock-price shock and the driver of volatility. The variance normalisation gives $\mathbb{E}[v_t] = \xi_0$.

The rough Bergomi specification follows Bayer, Friz and Gatheral [4]. The active simulator uses a hybrid discretisation of the Volterra integral: the near-diagonal Brownian integral is sampled using its conditional Gaussian law given the corresponding Brownian increment; the remaining kernel contributions are approximated using the hybrid scheme's evaluation weights. The previous joint-Cholesky implementation is retained as `rbergomi_paths_exact` for comparison. Stock prices are then advanced in log space with a left-point variance step. For the American put, the discounted reward is

$$R_j = e^{-rt_j}(K - S_{t_j})^+, \tag{13}$$

and the base features at each exercise date are S_{t_j}/K and v_{t_j}/ξ_0.

Simulator validation. The hybrid simulator passed the repository's existing 71-test suite; the additional P3 variance test also passed separately. A preliminary European-call check used $h = 0.07$, $\eta = 1.9$, $\rho = -0.9$, $\xi_0 = 0.0552$, $S_0 = K = 1$, $r = 0$, 50 time intervals and 30,000 paths. The Monte Carlo estimate was 0.07886 with standard error 0.00061, compared with the published reference value 0.0791. The difference is about 0.39 standard errors. This is a useful consistency check, not a high-precision benchmark reproduction, because the path count and time grid are smaller than those used for the reference calculation.

American put experiment (E4). We ran a reduced Hurst sweep at $h \in \{0.07, 0.10, 0.30, 0.50\}$ using 3,000 simulated paths, 25 time intervals and one random split per h. The model parameters were $T = 1$, $\eta = 1.9$, $\rho = -0.9$, $\xi_0 = 0.09$, $r = 0.05$, $S_0 = K = 100$. We compared Longstaff–Schwartz (LSM), Deep Optimal Stopping (DOS), OSPG-MLP and OSPG-GRU. The table reports the test-set mean reward and its pathwise standard error.

OSPG-MLP has the largest point estimate in these four pilot runs. However, the standard errors are around 0.31–0.37, and the uncertainty intervals overlap substantially. With only one split and 3,000 paths, these results do not establish that OSPG-MLP is superior to the other methods, nor

Table 4: Reduced E4 American put results under hybrid rough Bergomi. Entries are mean discounted payoffs (standard error).

h	LSM	DOS	OSPG-MLP	OSPG-GRU
0.07	7.935 (0.340)	7.985 (0.350)	8.136 (0.340)	7.932 (0.356)
0.10	7.698 (0.329)	7.776 (0.348)	7.934 (0.340)	7.690 (0.359)
0.30	7.362 (0.337)	7.435 (0.350)	7.507 (0.332)	7.367 (0.368)
0.50	7.540 (0.314)	7.619 (0.315)	7.605 (0.352)	7.479 (0.360)

do they establish a GRU advantage. The sweep is a preliminary computational comparison; more paths and independent splits would be needed for firm statistical conclusions.

4.3 Data and backtest methodology

We construct a long-only exit-timing experiment from daily historical market data. Each observation is a position entered in one asset on one session; a policy chooses its exit date, with mandatory liquidation by a fixed deadline. The unit of evaluation is a hypothetical trade window, rather than a self-financing multi-asset portfolio.

Sources and universe selection. Indian and US equities use Yahoo Finance daily closes with split and dividend adjustment explicitly enabled through `yfinance`. We freeze the NIFTY 50 list from the NSE and the S&P 100 constituent table at retrieval. Binance public spot daily klines supply crypto prices, with pagination, UTC daily sessions, and an exclusive end date. Crypto selection uses a frozen retrieval-date market-cap ranking from CoinGecko, excluding identified stablecoins and wrapped/staked representations. These are current-universe snapshots, not point-in-time historical constituents; survivorship and selection bias remain. Original requests are cached with source, retrieval date, symbols, adjustments and failures. Equity histories cover 2015–2024; crypto is requested from 2017 through 2024.

Market	Requested	Usable	Windows	Folds
NIFTY 50	50	50	20,014	12
S&P 100	101	100	48,444	12
Crypto	10	8	3,590	7

There are 101 requested US share-class tickers for the 100-company index. SNDK has no usable history in the frozen interval. Of ten requested crypto assets, FIGR_HELOC has no requested Binance spot pair and HYPE has no observations in the interval; eight are retained, including BTC and ETH. These exclusions are recorded, rather than replaced by simulated prices. Later listings contribute only after their own feature lookback is available. Missing terminal delisting proceeds are not imputed. FX and gold are omitted under the project's reduced-scope fallback; CSV/custom-symbol loaders support subsequent extensions with appropriate cost and currency assumptions.

Cleaning and calendars. We sort and deduplicate session dates, convert nonpositive/nonfinite prices to missing, exclude stablecoins, and log coverage and exclusions. Missing observations remain missing by default: there is no backward filling, interpolation, or extrapolation before listing or after the last observed price. Windows containing missing prices or features are discarded. Extreme log returns are flagged against preceding rolling statistics and retained, so a genuine crash is not removed as a data error. Equity sessions use observed dates within each local panel; crypto is reindexed to every UTC day. This does not reconstruct exchange-specific historical holidays or delisting cashflows.

Windows and observable features. The default deadline is 30 calendar days after entry, with exit on the last observed session on or before that deadline. Entries are spaced five panel observations apart. Windows without a complete horizon are excluded. Because calendar-month windows contain different numbers of sessions, arrays repeat the terminal state, reward and date up to 31 states; stopping indices are clipped to each window's true terminal index. Padding creates no additional price movement or holding opportunity.

Let S_0 be the entry reference price and S_j the current price. Features include $S_j/S_0 - 1$, running drawdown $S_j/\max_{k\le j} S_k - 1$, remaining time, the current daily log return, 20-observation trailing volatility, and log returns lagged by one, two and five observations. Volatility uses annualization factors 252 for equities and 365 for crypto. Local index and volatility-index levels (NIFTY/India VIX or S&P 500/VIX) are shifted one source observation before backward date alignment. This prevents using an external close that becomes available later than the decision. Feature construction is causal; any fitted scaling is learned on training data only.

Execution convention and costs. We use a stylized daily-close reference-price model. The current close and current exit reward are observable inputs, consistent with the existing stopping-method interface. We do not claim that an order submitted after observing a close can fill at that close. Next-open execution would require separating unobserved execution payoffs from policy inputs, which is outside this close-mark experiment.

Writing c_b, c_s for proportional entry/exit friction, f for a simple daily funding assumption and d_j for elapsed calendar days, the undiscounted net return is

$$r_j = \frac{S_j}{S_0}\frac{1-c_s}{1+c_b} - 1 - f d_j.$$

Both legs include fees, half-spread and slippage. Research fee scenarios per side are 12 basis points for NSE, one for US equities and ten for spot crypto; assumed half-spread/slippage are respectively 2/2, 1/1 and 2/2 basis points. These are explicit sensitivity assumptions, not official historical tariffs. Spot funding and discounting default to zero. A configurable perpetual-funding scenario represents an initial-notional charge, not a leveraged margin-account backtest. Cost multipliers and JSON overrides support sensitivity studies. The default reward is $R_j = r_j$; a second available objective is exponential utility $(1 - e^{-ar_j})/a$, with raw net returns saved separately.

Chronological evaluation and rules. Expanding walk-forward folds start with 36 months of training, six months validation and six months test, advancing by six months. All assets share date boundaries. A training or validation window is purged if its scheduled outcome horizon crosses the next boundary; an optional extra embargo is configurable. Test periods are disjoint. Trailing test features may use genuinely available earlier history. No threshold or model choice uses test outcomes. Fixed comparison rules are hold-to-deadline, take-profit/stop-loss at $+5\%/-3\%$, and a 5% trailing drawdown exit. Rules observe daily closes, rather than claiming intraday stop fills.

Validation and statistical handoff. The harness exports aligned asset IDs, unique window IDs, entry dates, all decision dates, deadlines, exit dates, realized rewards and raw net returns. All methods within a fold use identical test windows. Offline checks verify reward accounting, causal features and rules, boundary purging, missing-data behavior, calendar padding, cache replay, pagination, and metadata round trips. A real-data smoke run exercises every existing learned-method family with deliberately shortened neural training; it is an integration check, not the final learned-model comparison. Existing methods retain their global date-zero rule, originally designed for common simulated initial states; heterogeneous market entry states are a limitation. Overlapping windows and correlated assets invalidate independent-window standard errors. P5 receives dated per-path outputs for block-bootstrap inference, risk-sensitive comparisons and regime analysis. Mean trade returns are not interpreted as portfolio returns or annualized performance.

4.4 Backtest results

5 Conclusion and limitations

What we found. Optimal stopping can be solved without backward induction. The OSPG policy gradient optimises the expected reward of a stochastic stopping rule directly, using the exact distribution of its stopping time on pre-simulated paths; it needs neither rollouts nor the recursive continuation values on which the Snell-envelope methods rely. On the Markovian option benchmarks the memoryless OSPG policy comes closest to the exact prices, within 0.12% on the put and 0.44% on the 20-asset geometric call, ahead of Longstaff–Schwartz and fitted Q-iteration (Section 4.1); the binomial tree and our finite-difference solver of the free-boundary problem, which agree with each other to 4×10^{-4}, provide the exact prices. A recurrent policy pays a small price for memory it does not need (0.1–0.5% on these problems), but still beats the backward-induction methods where the problems are hard.

Limitations.

- All values are out of sample and therefore lower bounds; we report no upper bounds (dual methods), so the distance to the true price is known only where an exact price exists.
- For the 100-date geometric call we trained on 40,000 paths, about thirty times fewer than the paper's 1.28 million, and our errors there are correspondingly larger.
- OSPG needed three practical fixes (initial bias, logit penalty, separate date-0 decision) that the paper does not describe; without them it can converge to "stop immediately" or "never stop".
- The stopping rule is a hard threshold $\varphi_j \geq \frac{1}{2}$ on a stochastic policy trained for the soft objective; the two coincide only when the policy is nearly deterministic.
- The memory results rest on one rough-volatility model and one set of markets.

References

[1] N. Damera Venkata and C. Bhattacharyya. Deep recurrent optimal stopping. *NeurIPS*, 2023.

[2] S. Becker, P. Cheridito and A. Jentzen. Deep optimal stopping. *JMLR* 20(74), 2019.

[3] F. A. Longstaff and E. S. Schwartz. Valuing American options by simulation: a simple least-squares approach. *Review of Financial Studies* 14(1), 2001.

[4] C. Bayer, P. Friz and J. Gatheral. Pricing under rough volatility. *Quantitative Finance* 16(6), 2016.

[5] P. A. Forsyth and K. R. Vetzal. Quadratic convergence for valuing American options using a penalty method. *SIAM Journal on Scientific Computing* 23(6), 2002.

[6] S. E. Shreve. *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer, 2004.